



Survival Analysis in Cancer

Group No.20

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Introduction & Problem Statement

- In today's healthcare environment, **cancer remains one of the leading causes of mortality worldwide**, making outcome prediction extremely important.
- **Survival analysis** is a statistical technique used to **study the time until an event occurs**, such as death, relapse, or recovery in cancer patients.
- It helps researchers and doctors **understand patient survival patterns** and evaluate the effectiveness of different treatments.
- This project focuses on applying survival analysis techniques to cancer data to study patient survival time.
- The analysis considers key factors such as **age, cancer type, treatment method, and follow-up time**.
- The main objective is to **estimate survival probabilities**, compare treatment outcomes, and support **data-driven clinical decision-making**.

Solution of the problem

- To address challenges in understanding cancer outcomes and patient survival, this project provides:
- **Survival Time Estimation**
- **Handling Censored Data**
- **Comparison of Treatment Outcomes**
- **Risk Factor Analysis**

Features

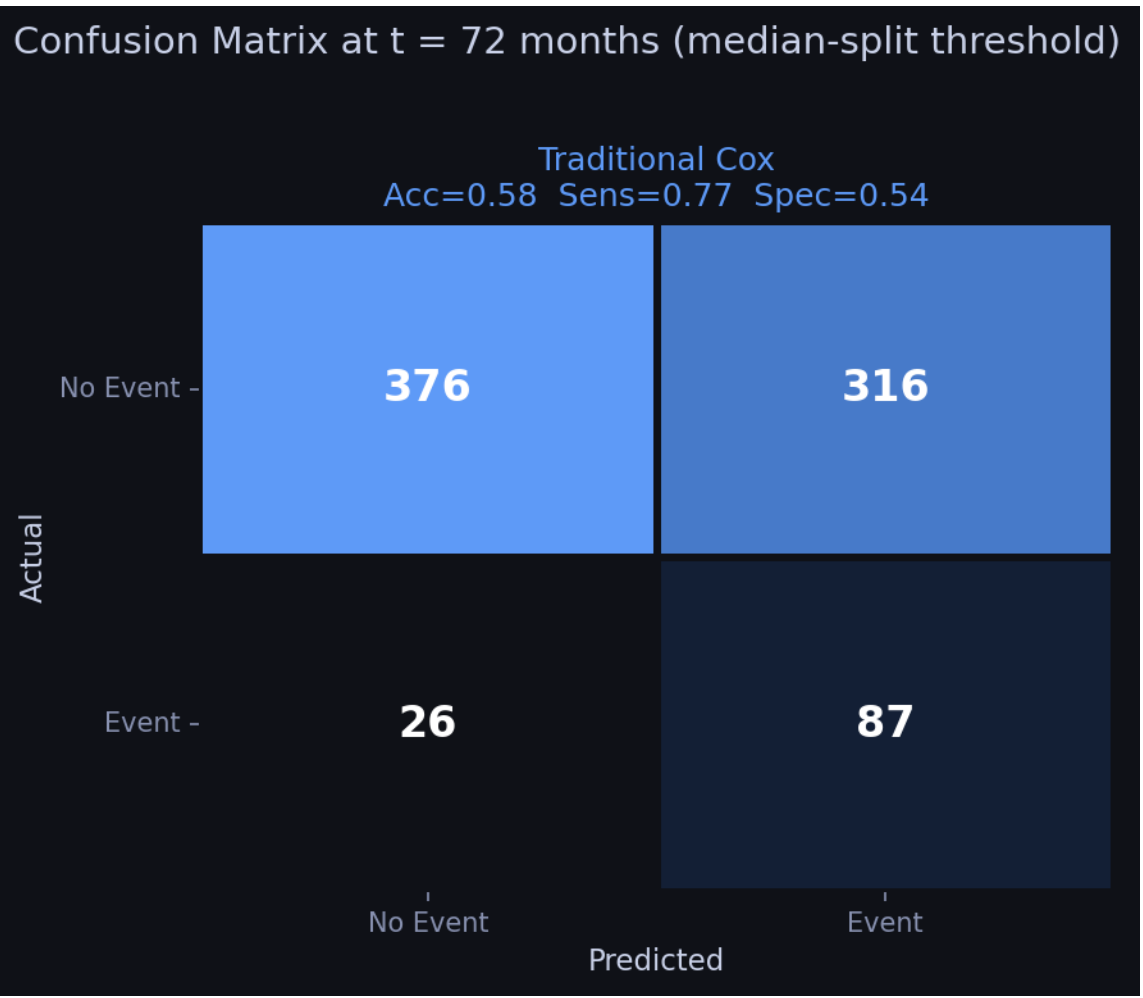
- 1. Accurate Survival Prediction**
- 2. Censored Data Handling**
- 3. Kaplan–Meier Survival Curves**
- 4. Treatment Comparison**
- 5. Risk Factor Evaluation**
- 6. Statistical Reliability**
- 7. Clinical Decision Support**
- 8. Easy Interpretation**

Project Scope

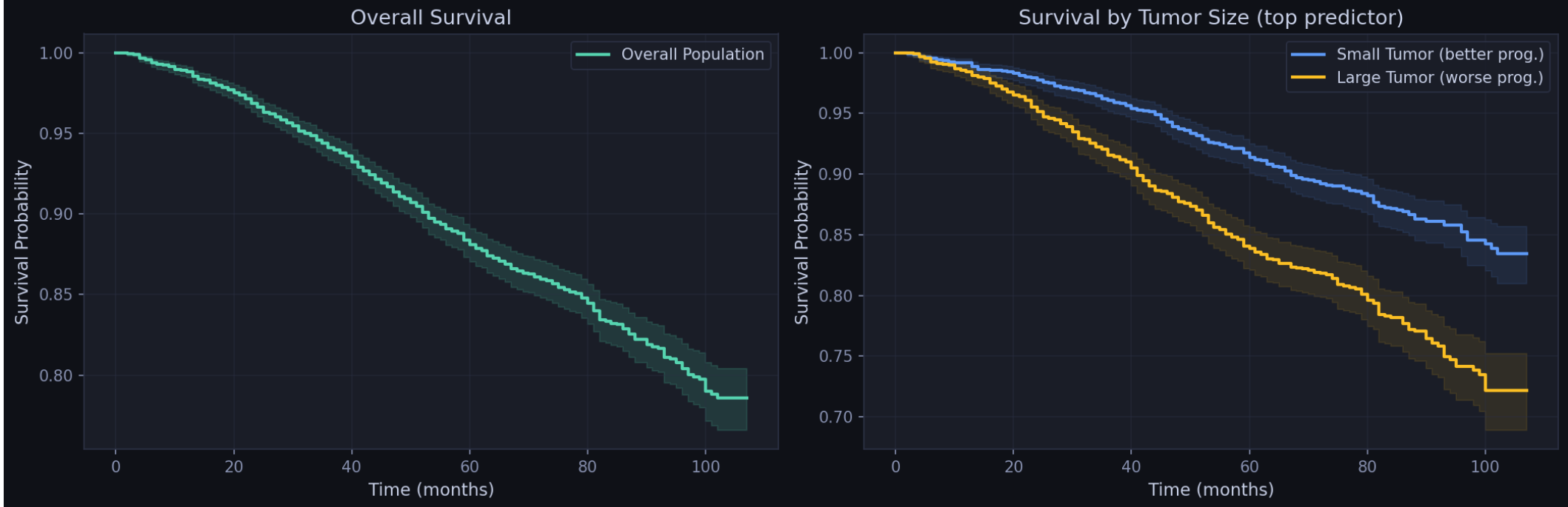
- Simple Analysis Approach
- Fast Data Processing
- Time-Saving System
- Cost-Effective Solution
- Well-Organized Data Handling
- Scalable for Research Use

Conclusion

- The project successfully demonstrates the application of **survival analysis techniques in cancer research**.
- It helps in **estimating patient survival time** and understanding survival patterns effectively.
- The analysis supports **comparison of treatment outcomes** and identification of key risk factors.
- The study highlights the importance of **statistical methods in healthcare decision-making**.
- Overall, the project shows how **data-driven survival analysis** can improve cancer research and clinical insights.



Kaplan-Meier Survival Curves • SEER Breast Cancer



THANK YOU!!!